COS 214 Practical 4

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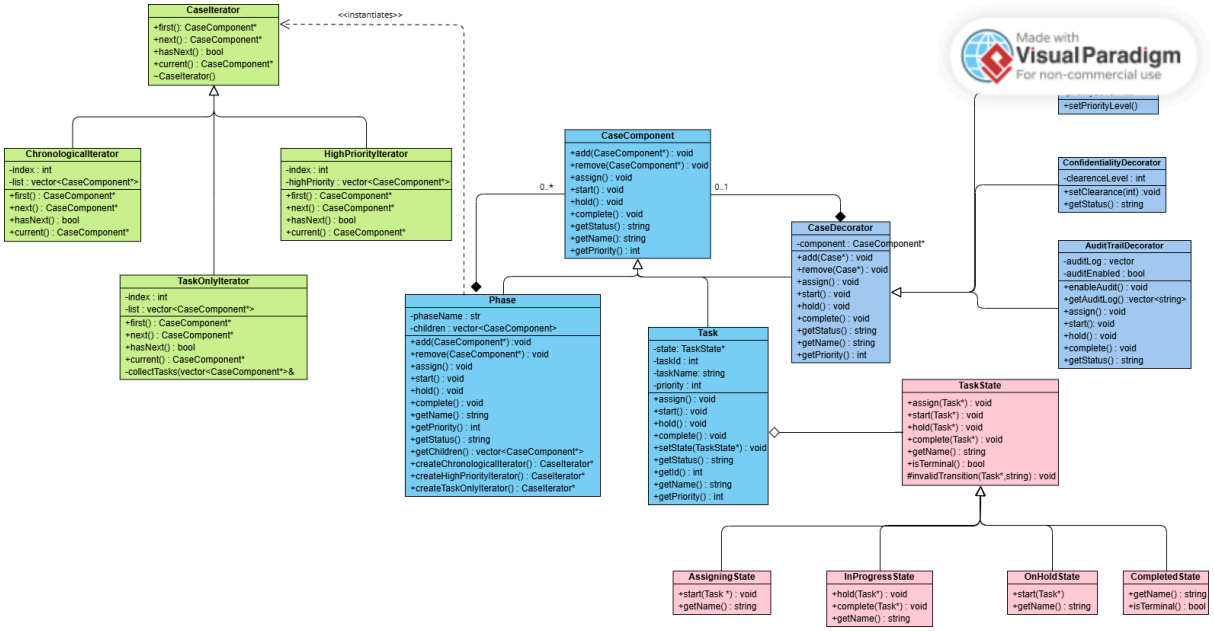
Github: https://github.com/yashGhela/COS214P4

Task 1:

UML Diagram:



Description:

We chose a case management system. It organizes work into Phases (containers) and Tasks (individual work items) that progress through a strict workflow lifecycle. The system provides multiple views of the investigation (chronological, task-only, and high-priority) and allows dynamic addition of audit trails and confidentiality controls to sensitive tasks. This gives investigation teams complete visibility, process compliance, and audit readiness.

GoF participants

Composite Pattern:

- Component: CaseComponent
- Leaf: Task
- Composite: Phase

State Pattern:

- Context: Task

- State: TaskState
- ConcreteState: AssigningState, InProgressState, OnHoldState, CompletedState

Decorator Pattern:

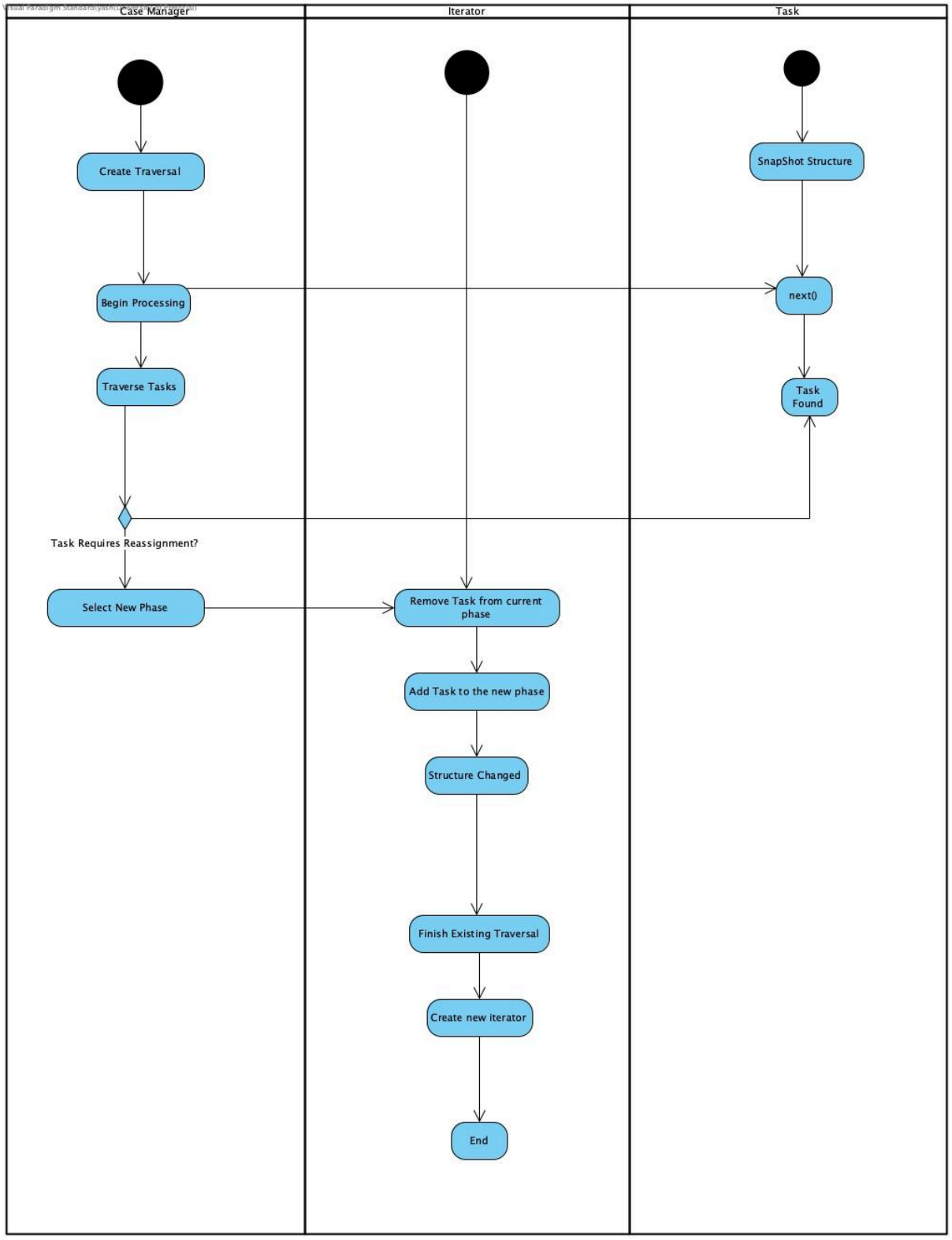
- Component: CaseComponent
- ConcreteComponent:Task
- Decorator: CaseDecorator
- ConcreteDecorator: AuditTrailDecorator, ConfidentialityDecorator, HighPriorityDecorator

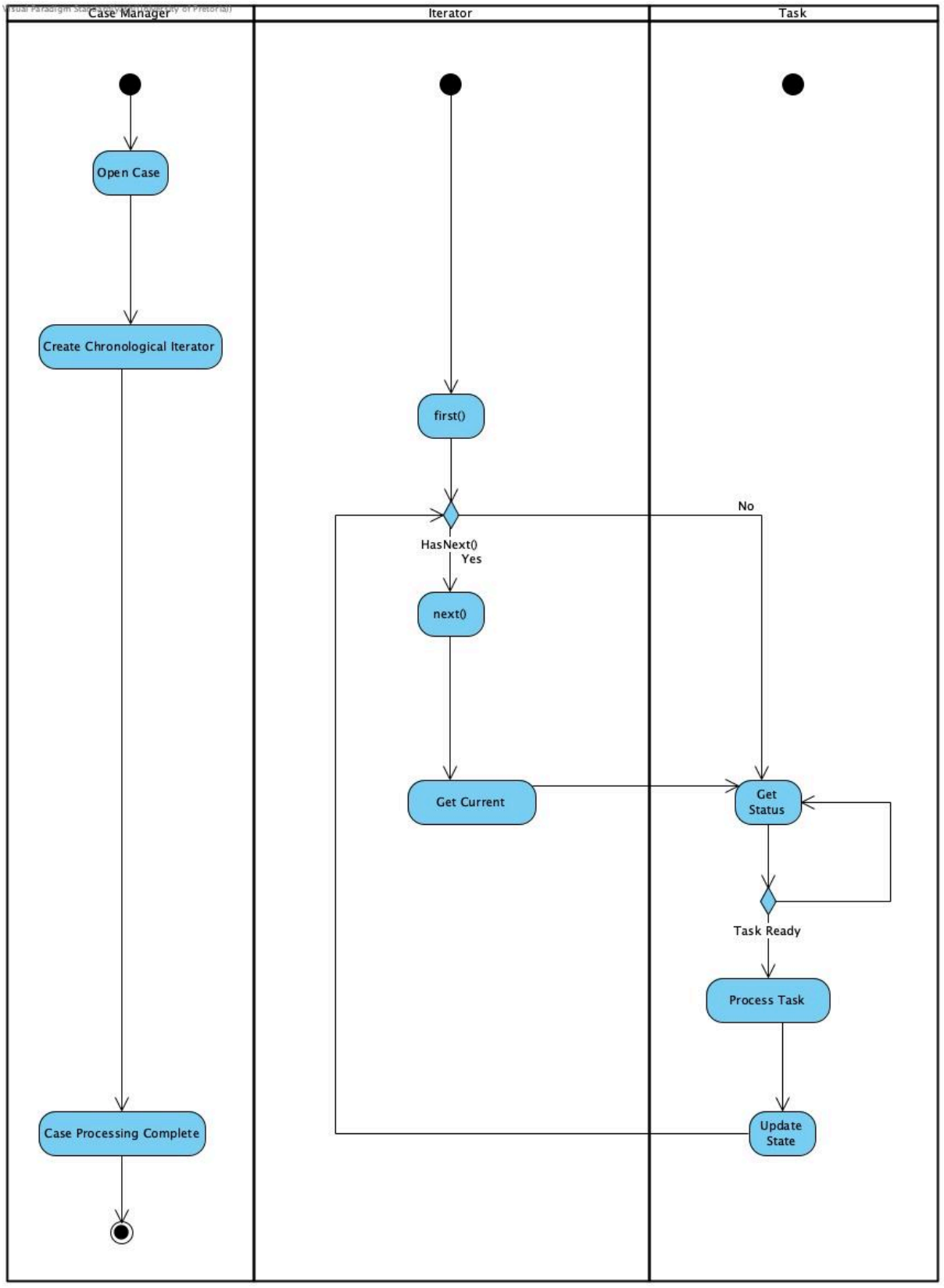
Iterator Pattern:

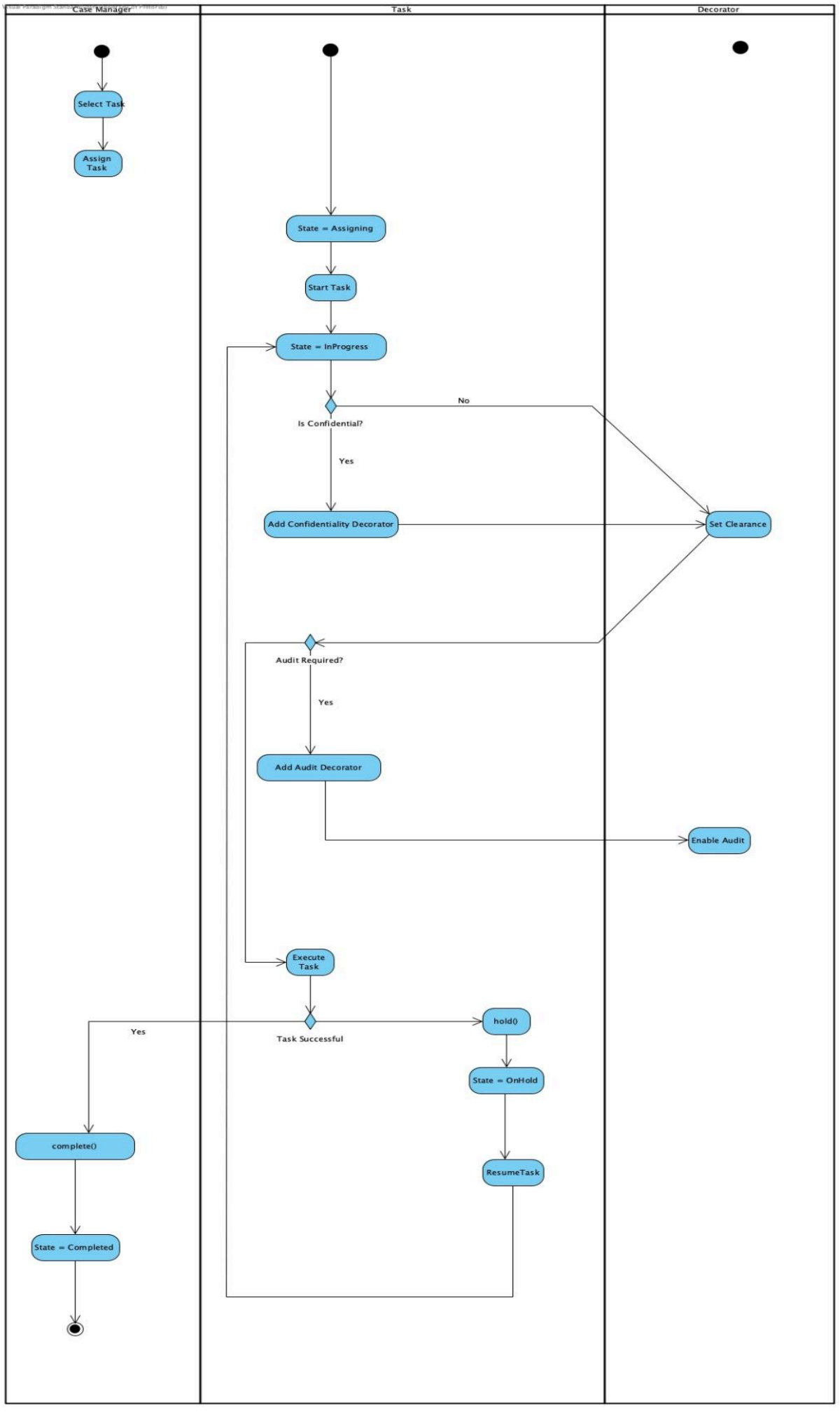
- Aggregate:Phase
- Iterator: CaseIterator
- ConcreteIterators : ChronologicalIterator, TaskOnlyIterator, HighPriorityIterator

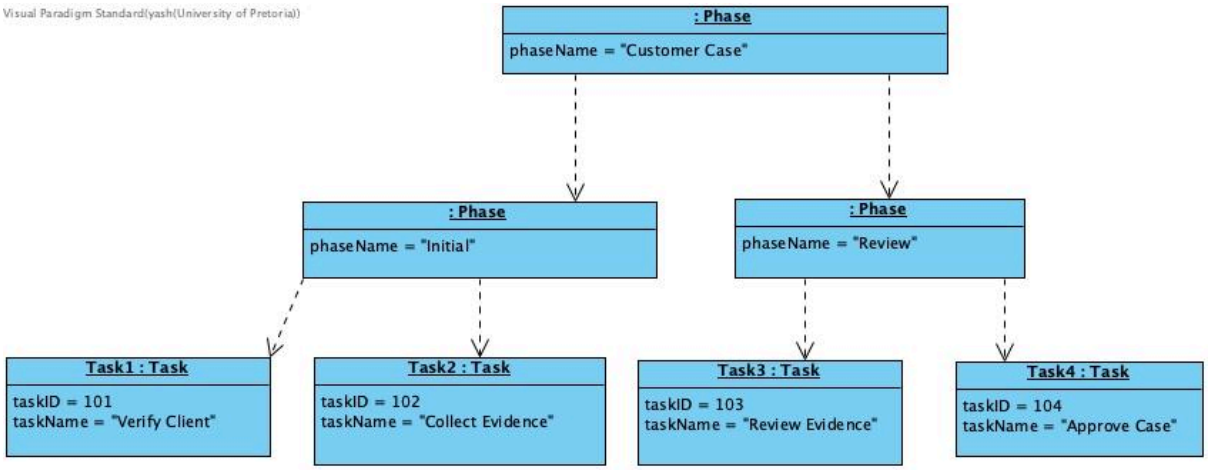
Ownership Rationale:
Phase owns its children (destructor deletes each one recursively, giving clean teardown of arbitrarily deep nesting), CaseDecorator owns the component it wraps (destructor deletes it, so deleting the outermost decorator cascades through the whole stack down to the underlying Task), and Task owns its current TaskState (deleted and replaced on every transition via setState). Iterators never own anything: they hold copies of pointers (a snapshot) for traversal, and are themselves owned and destroyed by whichever client created them.

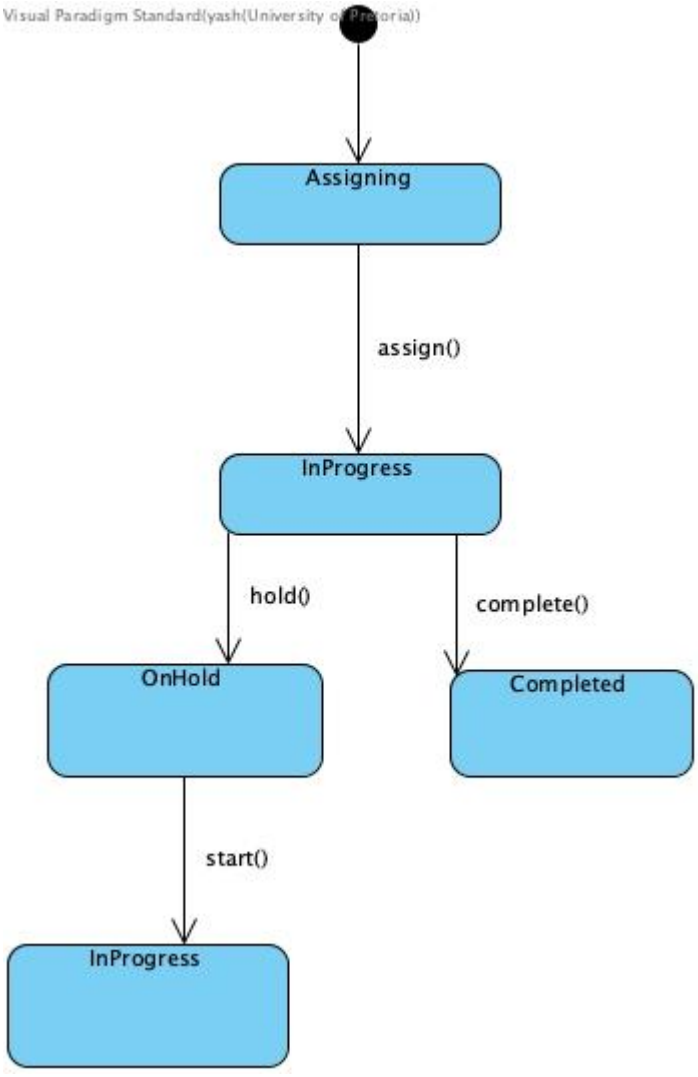
Task 4:


Case Manager
Iterator
Task
Create Traversal
Begin Processing
Traverse Tasks
Task Requires Reassignment?
Select New Phase
Remove Task from current phase
Add Task to the new phase
Structure Changed
Finish Existing Traversal
Create new iterator
End
SnapShot Structure
next()
Task Found


Case Manager
Iterator
Task
Open Case
Create Chronological Iterator
first()
HasNext()
No
Yes
next()
Get Current
Get Status
Task Ready
Process Task
Update State
Case Processing Complete


Case Manager
Task
Decorator
Select Task
Assign Task
State = Assigning
Start Task
State = InProgress
Is Confidential?
No
Yes
Add Confidentiality Decorator
Set Clearance
Audit Required?
Yes
Add Audit Decorator
Enable Audit
Execute Task
Task Successful
Yes
hold()
State = OnHold
ResumeTask
complete()
State = Completed

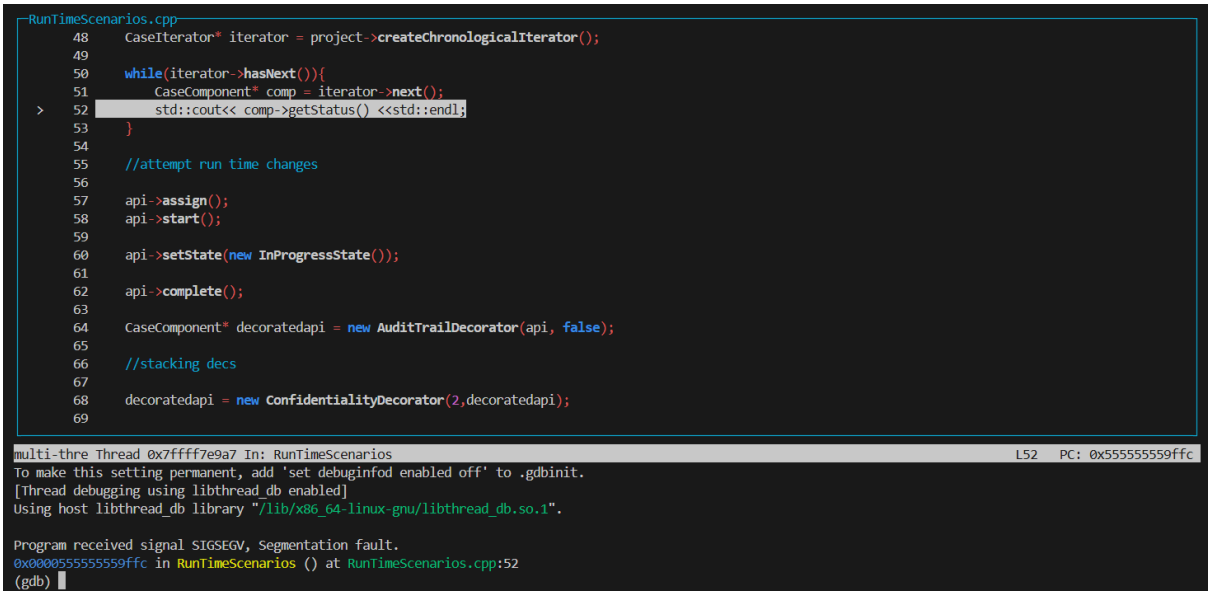




Task5:

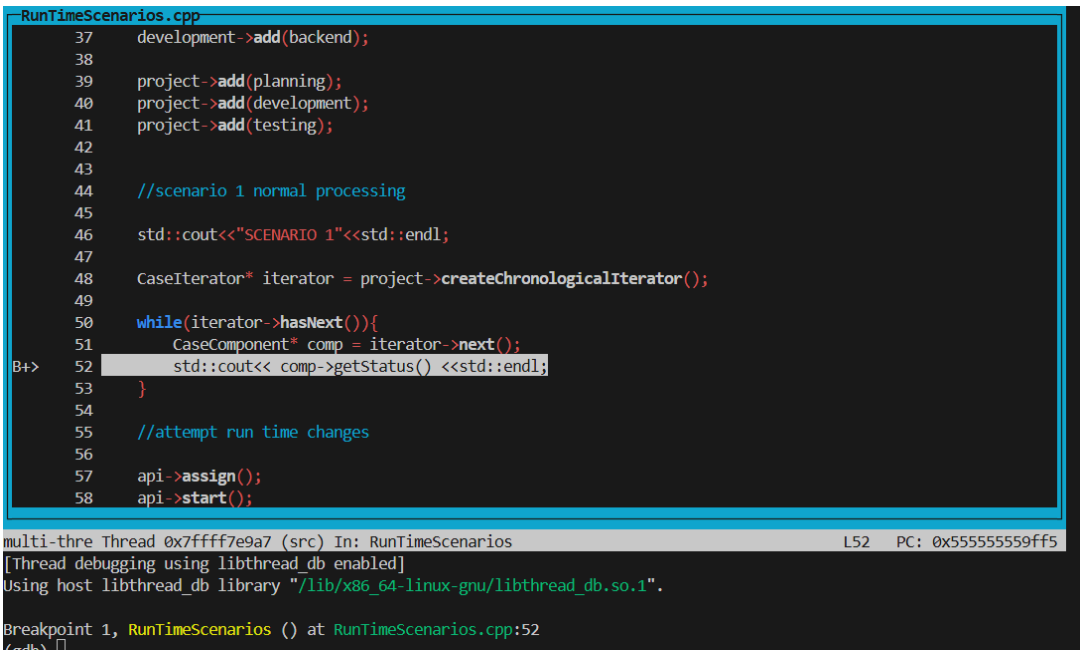
GDB Evidence:

Our runtimeScenarios function has caused a segmentation fault. The purpose of this function was to test different scenarios of ownership within our case system and the various state changes of tasks and how they behave.



Breakpoint usage:

We can assess that comp returns a nullptr, as the next function passes through its last element and returns a nullptr to getStatus causing the segfault to occur. So on the final iteration hasNext() is true before the call to next(). next() increments the index past the last element returning a nullptr and getStatus() ends up dereferencing that nullptr causing a segfault



The nullptr is after line 52:

```
Breakpoint 1, RunTimeScenarios () at RunTimeScenarios.cpp:52
(gdb) print comp
$5 = (CaseComponent *) 0x0
```

The fix: change in traversals
replace iterator->next() with iterator->current() returns the current index allowing getStatus to return the current status of the component within the list of the CaseComponents and by having iterator->next() after the getStatus() function call prevents nullptr from being returned. Finally ensuring to delete the iterator to ensure memory safety.

Debugging result post fix (Valgrind):

```
==57292==
==57292== HEAP SUMMARY:
==57292==     in use at exit: 0 bytes in 0 blocks
==57292==   total heap usage: 156 allocs, 156 frees, 79,187 bytes allocated
==57292==
==57292== All heap blocks were freed -- no leaks are possible
==57292==
==57292== For lists of detected and suppressed errors, rerun with: -s
==57292== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

TASK 6:

Teamwork Reflection:

The tasks were equally distributed amongst team members in regards to their availability and skillset. Thando had a focus on setting up or foundational design with Yash adding onto this systems design. There was a constant feedback loop between team members in regards to implementation and overall design structure. Tumisho had a heavy basis on code testing and deployment. A great defining commit would be Thando's initial commit of the base implementation of the system, allowing Tumisho and Yash to better grasp the implementation of the design patterns and how the different components of the Case system relate and interact with each other. Yash's runtime scenarios commit enabled us to see the system in use before an official demoing main.